

RIVIERE-DU-LOUP, QC, Canada

WMO: 717150

Lat: 47.8061N

Lon: 69.5486W

Elev: 147

StdP: 99.58

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -24.1	(c) -21.7	(d) -30.5	(e) 0.2	(f) -23.4	(g) -27.7	(h) 0.3	(i) -21.0	(j) 9.3	(k) -4.8	(l) 8.2	(m) -6.6	(n) 2.9	(o) 190	(p) 0.588

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.8	26.3	19.8	24.8	19.0	23.4	18.2	21.1	24.7	20.1	23.4	19.1	22.3	3.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.8	14.8	23.3	18.8	13.9	22.3	17.8	13.0	21.3	62.0	24.7	58.5	23.4	55.0	22.5	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.0	6.9	6.1	-26.5	29.6	2.2	1.7	-28.1	30.8	-29.4	31.8	-30.6	32.8	-32.2	34.0		
			-27.0	23.2	2.2	1.2	-28.5	24.0	-29.8	24.7	-31.0	25.4	-32.6	26.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.9	-11.5	-9.9	-4.7	2.2	9.3	14.6	17.8	17.0	12.6	6.1	-0.2	-7.0
	DBStd	11.10	6.43	5.88	5.40	3.91	3.97	3.77	3.07	3.11	3.82	4.03	4.64	5.51
	HDD10.0	2968	667	557	458	236	62	4	0	0	16	136	306	527
	HDD18.3	5330	925	790	716	485	283	123	47	64	177	380	555	786
	CDD10.0	750	0	0	1	1	39	143	242	217	92	14	1	0
	CDD18.3	69	0	0	0	0	1	12	31	22	4	0	0	0
	CDH23.3	323	0	0	0	0	14	58	142	90	18	0	0	0
	CDH26.7	37	0	0	0	0	1	7	19	8	1	0	0	0
Wind	WSAvg	2.7	3.1	3.1	3.2	3.0	2.8	2.4	2.2	2.2	2.4	2.7	2.9	3.0
Precipitation	PrecAvg	977	68	64	72	74	88	84	100	98	86	87	76	94
	PrecMax	1223	134	118	116	176	183	159	264	306	194	185	136	321
	PrecMin	757	18	32	30	29	29	25	21	35	50	24	28	32
	PrecStd	117	29	22	24	38	36	33	62	51	31	42	26	55
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.0	6.2	10.2	16.8	25.4	27.5	28.8	27.8	25.5	19.7	15.6	6.9
		MCWB	6.3	4.0	6.2	10.7	16.6	19.9	21.3	20.7	20.0	15.3	12.7	5.9
	2%	DB	3.1	2.4	6.2	12.7	21.7	24.9	26.5	25.6	23.0	17.3	10.9	3.5
		MCWB	2.3	0.8	3.2	7.3	14.6	18.7	20.3	19.6	18.4	13.8	8.5	2.5
	5%	DB	0.2	0.5	4.2	10.0	19.0	23.0	24.8	24.1	21.0	14.7	8.4	1.8
		MCWB	-0.5	-0.5	1.7	5.8	13.3	17.4	19.3	18.7	17.1	11.8	6.4	0.7
	10%	DB	-2.3	-1.2	2.4	7.9	16.4	21.3	23.3	22.6	19.1	12.3	6.0	0.4
		MCWB	-3.1	-2.1	0.4	4.5	11.3	16.4	18.5	18.1	15.7	9.8	4.2	-0.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.4	4.7	7.6	12.0	18.4	21.8	23.0	21.8	20.9	16.3	13.4	6.3
		MCDB	7.0	5.7	9.2	15.2	23.5	26.1	27.2	25.9	24.4	18.9	14.6	7.0
	2%	WB	2.4	1.2	3.6	8.0	16.1	19.8	21.4	20.6	19.1	14.1	9.1	2.7
		MCDB	3.0	2.1	5.5	11.3	20.2	23.2	25.0	24.1	22.0	16.7	10.4	3.4
	5%	WB	-0.5	-0.4	2.0	6.3	13.7	18.5	20.3	19.7	17.6	12.2	6.5	0.9
		MCDB	0.2	0.4	3.8	9.5	17.6	21.9	23.5	23.0	20.5	14.5	8.1	1.6
	10%	WB	-3.1	-2.0	0.6	4.7	11.8	17.1	19.2	18.7	16.0	10.1	4.5	-0.4
		MCDB	-2.4	-1.2	2.2	7.3	15.7	20.4	22.4	21.8	18.7	12.0	5.8	0.3
Mean Daily Temperature Range	5% DB	MDBR	7.6	7.6	7.2	7.2	9.2	9.3	8.8	8.8	8.4	6.5	5.6	6.1
		MCDBR	9.5	7.7	8.4	11.1	13.2	11.2	10.5	10.0	10.3	9.5	8.6	7.2
		MCWBR	9.5	7.1	6.8	7.7	7.9	7.0	6.0	5.8	7.0	7.1	7.7	7.1
	5% WB	MCDBR	9.6	7.3	7.6	10.0	11.9	9.9	9.5	9.2	10.2	8.9	8.6	7.2
		MCWBR	9.6	6.9	6.6	7.5	8.0	7.1	6.4	6.2	7.7	7.3	7.9	7.2
Clear-Sky Solar Irradiance	taub	0.267	0.275	0.312	0.354	0.360	0.378	0.405	0.385	0.361	0.328	0.294	0.271	
	taud	2.319	2.287	2.250	2.306	2.368	2.364	2.364	2.310	2.374	2.433	2.513	2.476	2.359
	Ebn at Noon	816	891	905	897	902	884	853	857	846	826	784	763	
	Edh at Noon	71	92	113	119	117	118	123	111	95	75	63	61	
All-Sky Solar Radiation	RadAvg	1.31	2.23	3.43	4.44	5.08	5.46	5.43	4.71	3.64	2.13	1.16	0.89	
	RadStd	0.16	0.26	0.26	0.36	0.51	0.37	0.53	0.31	0.26	0.19	0.13	0.09	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page